

Exercise - 02

Aim: Study of different types of Network cables

Cable type	Category	Maximum Data Transmission	Advantages/Dis-advantages	Application	Image
UTP	Category 3	10 bps	Advantages • cheaper in cost	10 Base-T Ethernet	 UTP Shielded Pairs with colour coded plastic insulation
	Category 5	upto 100 mbps	• Easy to install as they have a smaller overall diameter	Fast Ethernet Gigabit Ethernet Ethernet	
	Category 5e	1 Gbps	Disadvantages • More prone to (EMI) Electromagnetic Interference and noise	Fast Ethernet Gigabit Ethernet	
STP	Category 6, 6a	10 Gbps	Advantages • Shielded • Faster than UTP • Less susceptible to noise and interference	Gigabit Ethernet, 10Gb Ethernet (55m) widely used in data centres	 UTP Shielded Pair shield twisted pair cables
SSTP	Category 7	10 Gbps	Disadvantage • Expensive • Greater installation effort	Gigabit Ethernet, 10G Ethernet (100m)	

Coaxial
Cable

RG-8
RG-59
RG-11

10-100
Mbps

- Advantages
- High bandwidth
 - Immune to Interference
 - Low loss bandwidth
 - Versatile
- Disadvantages

- Limited distance
- cost
- size is bulky

Speed of
Signal is
soon
Television
network
high
Speed
Internet
Connection



Fibre
Optics
Cable

Single-
mode
Multi-
mode

100 Gbps

- Advantages
- High Speed
 - High bandwidth
 - High Security
 - Long distance
- Disadvantage
- Expensive
 - Requires skilled installers

Maximum
distance of Fibre
Optics
Cable is
around
100
meters



Result: Study of different types of Network cables is